PHYS1408 Practice Exam 2-7 — Answer Key

Worked solutions

All answers were checked numerically with `_scripts/build_exam_2_7.py`. $g = 9.8$ m/s^2 throughout.

Practice Exam 2-7 is the **2015 Exam 2, question by question, with new numbers**. It keeps the same wording, the same unknowns and the same style of answer choices, including his evenly spaced choices on #1, #9 and #12 and the percentage ranges on #7 and #8. Questions 14 and 15 are lecture-note examples (Atwood's machine; α from an angle). Question 3 is a **minimum** trap: round up. The last column says which 3×5 notecard entry solves each question.

Q	Answer	Value	Topic	Notecard entry
1	**d**	4.4 m	Halfpipe with a friction patch	Halfpipe
2	**c**	14.2 m	Projectile max height from a ledge	Projectile
3	**e**	24.2 m/s → 26	Loop-the-loop, minimum speed at the bottom	Loop
4	**a**	0.97 s	Sliding up a rough ramp, time to stop	One block on incline (Stop up, Time)
5	**c**	1.7 m	Vertical spring launch	Spring (launch up)
6	**b**	+2.6 m/s	1-D elastic collision	Elastic
7	**e**	94%	Inelastic collision, KE fraction (symbolic)	Stick (inelastic)
8	**b**	0.13%	Ballistic pendulum, KE fraction	Ballistic
9	**d**	2.33 m/s^2	Hanging mass on a pulley with I	Atwood / mass on pulley (one mass)
10	**b**	229 rad/s	Rolling sphere, find ω	Rolling (sphere row)
11	**e**	1.85 rad/s → +2	Counter-rotating wheel dropped on a disk	Angular momentum
12	**d**	0.93 rad/s	Child jumps onto a merry-go-round	Angular momentum
13	**a**	2.05 m/s	Incline + friction + disk pulley	Incline + hanging mass + pulley ($m_2 = 0$)

Q	Answer	Value	Topic	Notecard entry
14	**a**	2.42 m/s	Atwood's machine	Atwood / mass on pulley
15	**c**	16 rad/s^2	Rotational kinematics, α from an angle	Rot. kinematics

1. Halfpipe with a friction patch — d) 4.4 m

Source: 2015 Exam 2 #1; Ch 7/8 notes p. 4/18–5/18. **Notecard:** Halfpipe.

Notecard, starting from rest with one crossing ($v_i = 0$, $n = 1$):

$$h_f = h_i - \mu L$$

Plug in:

$$h_f = 5.0 - (0.15)(4.0) = 5.0 - 0.60 = 4.4 \text{ m}$$

Distractors:

- a)–c) and e) are 0.1 m steps around the answer, like the 2015 exam's choices. Only $h_i - \mu L$ gives 4.4.

2. Projectile max height from a ledge — c) 14.2 m

Source: 2015 Exam 2 #2. **Notecard:** Projectile.

Notecard:

$$H = h_0 + \frac{(v_0 \sin\theta)^2}{2g}$$

Plug in ($v_0 \sin\theta = 22 \sin 40^\circ = 14.14$ m/s):

$$H = 4.0 + \frac{(14.14)^2}{2(9.8)} = 4.0 + \frac{200}{19.6} = 4.0 + 10.2 = 14.2 \text{ m}$$

Distractors:

- a) 4.0 is just the ledge.
- b) 10.2 forgets to add the ledge.
- d) 18.5 uses $\cos\theta$.
- e) 28.7 uses all of v_0 (as if thrown straight up).

3. Loop-the-loop, minimum speed at the bottom — e) 26 m/s

Source: 2015 Exam 2 #3; Ch 7/8 notes p. 12/18–13/18. **Notecard:** Loop.

Notecard: bottom minimum $v_b = \sqrt{5gR}$.

Plug in:

$$v_b = \sqrt{5(9.8)(12)} = \sqrt{588} = 24.2 \text{ m/s}$$

Answer-choice trap. 24.2 m/s is a **minimum**. At 24 m/s the mass falls off the loop, even though 24 is the closest number. The smallest choice that works is **e) 26 m/s**.

Distractors:

- a) 11 is $\sqrt{gR}$, the minimum at the **top**.
- b) 19 is $\sqrt{3gR}$.
- c) 22 is $\sqrt{4gR}$: just enough to reach the top with zero speed.
- d) 24 is the nearest number, but it's below the minimum.

4. Sliding up a rough ramp, time to stop — a) 0.97 s

Source: 2015 Exam 2 #4. **Notecard:** One block on incline (Stop up, Time).

Step 1: Stopping distance (notecard, stop up):

$$\Delta x = \frac{v_i^2}{2g(\sin\theta + \mu\cos\theta)} = \frac{6.0^2}{2(9.8)(0.1392 + 0.50 \times 0.9903)} = \frac{36}{19.6(0.6343)} = 2.90 \text{ m}$$

Step 2: Time (notecard, $v_f = 0$):

$$t = \frac{2\,\Delta x}{v_i + v_f} = \frac{2(2.90)}{6.0 + 0} = 0.97 \text{ s}$$

The 4.0 kg mass cancels.

Distractors:

- b) 1.72 uses the sliding-**down** bracket, $(\mu\cos\theta - \sin\theta)$.
- c) 1.93 has an extra factor of 2.
- d) 2.90 is the stopping **distance** in meters, not the time.
- e) 4.40 ignores friction.

5. Vertical spring launch — c) 1.7 m

Source: 2015 Exam 2 #5; Ch 7/8 notes p. 12/18. **Notecard:** Spring (launch up).

Notecard, launch up (measured from the release point):

$$h = \frac{kx^2}{2mg}$$

Plug in:

$$h = \frac{(1500)(0.030)^2}{2(0.040)(9.8)} = \frac{1.35}{0.784} = 1.7 \text{ m}$$

Distractors:

- a) 0.17 slips a decimal.
- b) 0.86 has an extra factor of 2 in the denominator.
- d) 3.4 forgets the $\frac{1}{2}$ on the spring PE.
- e) 17 slips a decimal the other way.

6. 1-D elastic collision — b) +2.6 m/s

Source: 2015 Exam 2 #6; Ch 9 notes p. 3/5. **Notecard:** Elastic.

Notecard formulas, right = +: $m_1 = 2.0$, $v_{1i} = +4.0$; $m_2 = 3.0$, $v_{2i} = -3.0$.

$$v_{1f} = \frac{(m_1 - m_2)v_{1i} + 2m_2 v_{2i}}{m_1 + m_2} = \frac{(-1.0)(4.0) + 2(3.0)(-3.0)}{5.0} = \frac{-4 - 18}{5} = -4.4 \text{ m/s}$$

$$v_{2f} = v_{1i} + v_{1f} - v_{2i} = 4.0 + (-4.4) - (-3.0) = +2.6 \text{ m/s}$$

Distractors:

- a) +4.4 is the size of the 2.0 kg mass's velocity.
- c) −0.2 is the stick-together answer.
- d) −2.6 has the sign wrong.
- e) −4.4 is the 2.0 kg mass's velocity.

7. Inelastic collision, KE fraction (symbolic) — e) >90%

Source: 2015 Exam 2 #7; Ch 9 notes p. 1/5. **Notecard:** Stick (inelastic).

Symbolic, so **set** $m = 1$ **and** $v = 1$: $m_1 = 3$, $v_1 = 2$; $m_2 = 1$, $v_2 = 1$.

Shared velocity (notecard):

$$v_f = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2} = \frac{3(2) + 1(1)}{4} = 1.75$$

KE fraction (notecard):

$$\frac{K_f}{K_i} = \frac{(m_1 + m_2) v_f^2}{m_1 v_1^2 + m_2 v_2^2} = \frac{4(1.75)^2}{3(2)^2 + 1(1)^2} = \frac{12.25}{13} = 0.94$$

94% is retained: **>90%**. (When both cars move the same way at similar speeds, little KE is lost.)

Distractors:

- d) 30–50% would need a big speed difference.
- a) <1% is the ballistic-pendulum size of loss.

8. Ballistic pendulum, KE fraction — b) 0.1–0.5%

Source: 2015 Exam 2 #8; Ch 9 notes p. 5/5. **Notecard:** Ballistic.

Notecard: $\frac{K_f}{K_i} = \frac{m}{M + m}$. The swing height isn't needed.

$$\frac{K_f}{K_i} = \frac{0.008}{6.0 + 0.008} = 0.0013 = 0.13\%$$

That's in the **0.1–0.5%** range.

Distractors:

- a) <0.1% would need a block heavier than 8 kg.
- e) 10% would mean the collision kept a lot of KE, which a sticking bullet never does.

9. Hanging mass on a pulley with I — d) 2.33 m/s²

Source: 2015 Exam 2 #9; Ch 10 notes p. 4/8. **Notecard:** Atwood / mass on pulley (one mass).

Notecard, one mass: $a = \dfrac{mg}{m + I/R^2}$. Here $I = \frac{3}{5}MR^2$, so $I/R^2 = \frac{3}{5}M = 4.8$ kg (R cancels).

$$a = \frac{mg}{m + \frac{3}{5}M}$$

Plug in:

$$a = \frac{(1.5)(9.8)}{1.5 + 4.8} = \frac{14.7}{6.3} = 2.33 \text{ m/s}^2$$

Distractors:

- The other choices are 0.5 m/s² steps around the answer, like the 2015 exam's. $R = 0.20$ m is a distractor: it cancels.

10. Rolling sphere, find ω — b) 229 rad/s

Source: 2015 Exam 2 #10; Ch 10 notes p. 6/8–8/8. **Notecard:** Rolling (sphere row).

Notecard, sphere: $v = \sqrt{\frac{10}{7}gh}$, then $\omega = \dfrac{v}{R}$.

$$v = \sqrt{\tfrac{10}{7}(9.8)(1.5)} = \sqrt{21} = 4.58 \text{ m/s}$$

$$\omega = \frac{v}{R} = \frac{4.58}{0.020} = 229 \text{ rad/s}$$

Distractors:

- a) 4.6 is v (m/s), not ω.
- c) 458 divides by $R/2$.
- d) 2290 slips a decimal in R.
- e) 52,500 is ω^2.

11. Counter-rotating wheel dropped on a disk — e) +2 rad/s

Source: 2015 Exam 2 #11; Ch 11 notes p. 3/8. **Notecard:** Angular momentum.

Notecard, with the wheel's ω_2 **negative**:

$$\omega_f = \frac{I_1\omega_1 + I_2\omega_2}{I_1 + I_2}$$

Moments of inertia first:

$$I_1 = \tfrac{1}{2}(150)(2.0)^2 = 300 \text{ kg}\cdot\text{m}^2$$

$$I_2 = 0.8(50)(1.5)^2 = 90 \text{ kg}\cdot\text{m}^2$$

Plug in:

$$\omega_f = \frac{(300)(3.0) + (90)(-2.0)}{300 + 90} = \frac{900 - 180}{390} = 1.85 \text{ rad/s}$$

The nearest choice is **e) +2 rad/s**, the same as on the 2015 exam.

Distractors:

- d) +1.5 is the other neighbor, but 1.85 is closer to 2.
- a) and b) come from adding the wheel's I without its spin, or flipping signs.

12. Child jumps onto a merry-go-round — d) 0.9 rad/s

Source: 2015 Exam 2 #12; Ch 11 notes p. 4/8. **Notecard:** Angular momentum.

Notecard with $+mvR$ (same direction) and $I_{add} = mR^2$ (person at the rim):

$$\omega_f = \frac{\frac{1}{2}MR^2\,\omega_i + mvR}{\frac{1}{2}MR^2 + mR^2}$$

Plug in ($\frac{1}{2}MR^2 = \frac{1}{2}(400)(2.5)^2 = 1250 \text{ kg}\cdot\text{m}^2$):

$$\omega_f = \frac{(1250)(0.60) + (45)(6.0)(2.5)}{1250 + (45)(2.5)^2}$$

$$= \frac{750 + 675}{1250 + 281} = \frac{1425}{1531} = 0.93 \text{ rad/s}$$

The nearest choice is **d) 0.9 rad/s**.

Distractors:

- b) 0.5 ignores the child's running momentum (0.49).
- e) 1.1 forgets to add the child's mR^2 (1.14).

13. Incline + friction + disk pulley — a) 2.05 m/s

Source: 2015 Exam 2 #13. **Notecard:** Incline + hanging mass + pulley ($m_2 = 0$).

Notecard with no hanging mass ($m_2 = 0$), block sliding **down**:

$$v_f = \sqrt{v_i^2 + \frac{2g\,\Delta x\,B}{m + I/R^2}}$$

$$B = m\sin\theta - \mu m\cos\theta$$

Net pull ($\sin 25^\circ = 0.4226$, $\cos 25^\circ = 0.9063$):

$$B = 6.0(0.4226) - 0.12(6.0)(0.9063) = 2.536 - 0.653 = 1.883 \text{ kg}$$

Pulley term: $I = 0.5MR^2$, so $I/R^2 = 0.5M = 1.5$ kg.

Plug in:

$$v_f = \sqrt{0.50^2 + \frac{2(9.8)(0.80)(1.883)}{6.0 + 1.5}}$$

$$= \sqrt{0.25 + \frac{29.53}{7.5}} = \sqrt{0.25 + 3.94} = 2.05 \text{ m/s}$$

Distractors:

- b) 1.98 forgets v_i^2.
- c) 1.88 treats the pulley as a ring ($I/R^2 = M$).
- d) 1.49 forgets the 2 in $2g$.
- e) $B > 0$, so it speeds up and never stops.

14. Atwood's machine — a) 2.42 m/s

Source: Ch 7/8 notes p. 2/18. **Notecard:** Atwood / mass on pulley.

Notecard, massless pulley ($I = 0$), from rest:

$$v_f = \sqrt{\frac{2gh(m_1 - m_2)}{m_1 + m_2}}$$

Plug in:

$$v_f = \sqrt{\frac{2(9.8)(1.5)(12 - 8.0)}{12 + 8.0}} = \sqrt{\frac{117.6}{20}} = \sqrt{5.88} = 2.42 \text{ m/s}$$

Distractors:

- b) 3.13 divides by m_1 only.

- c) 3.83 divides by m_2 only.
- d) 4.85 doubles the answer.
- e) 5.42 is free fall, $\sqrt{2gh}$.

15. Rotational kinematics, α from an angle — c) 16 rad/s²

Source: Ch 11 notes p. 6/8. **Notecard:** Rot. kinematics.

No time is given or asked for, so use the equation without t:

$$\omega_f^2 = \omega_i^2 + 2\alpha\,\Delta\theta$$

Solve for α:

$$\alpha = \frac{\omega_f^2 - \omega_i^2}{2\,\Delta\theta}$$

Plug in:

$$\alpha = \frac{60^2 - 20^2}{2(100)} = \frac{3600 - 400}{200} = 16 \text{ rad/s}^2$$

Distractors:

- a) 0.40 is $\Delta\omega/\Delta\theta$.
- b) 8.0 squares the difference: $(\omega_f - \omega_i)^2/2\Delta\theta$.
- d) 32 forgets the 2.
- e) 36 drops ω_i and the 2.